Group II Row IV (full cross section): DGLAP, running coupling and finite terms, step by step

1 Result

Corrected version. The earlier version read your $\frac{\xi}{(1-\xi)_+}$ as a plus distribution in ξ. It is your p^+ prescription, $\int dp^+ \frac{F(p^+)-F(0)}{p^+}$, and only that reproduces your (L2) exactly (Sec. 3). With it, the $\log \frac{\vee}{k^+}\,\mathrm{LO}[\mathcal{B}]$ of (L2) cancels; it is *not* a BFKL term. Exactly, up to $O(\Lambda/k^+)$:

$$\begin{aligned}(\mathrm{L1})+(\mathrm{L2})+(\mathrm{L3}) = {} & \underbrace{\frac{g^2N_c}{4\pi^2}\log\frac{k^+}{\Lambda}\,\mathrm{LO}[\mathcal{B}]}_{\text{JIMWLK list}} \\ & + \underbrace{\Big(\text{pole}\to D_{h/g}\Big) - \frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\mathrm{LO}_\xi\Big[\log\frac{(y'-y)^2\mu_F^2}{c_0^2}\Big]}_{\text{DGLAP}} \\ & + \underbrace{\frac{g^2\beta_0}{16\pi^2}\,\mathrm{LO}\Big[\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big]}_{\text{RC}} + \underbrace{\frac{g^2N_c}{4\pi^2}\int_a^1 d\xi\,\frac{1-\xi}{\xi}\,\mathrm{LO}^{\mathrm{lp}}_\xi\Big|_{d_\perp=2}}_{\text{finite}}\end{aligned} \tag{1.1}$$

Notation is in Sec. 2: $a=\frac{k^+}{\vee}$, $\mathcal{B}=-\frac{2}{\epsilon}-\frac{d_\perp}{2}\ell_\Delta(\mu)$; the running coupling is after the coupling counterterm.

- **The DGLAP term contains** $+\log\frac{\vee}{k^+}H_4$ at its lower end $\xi\to a$ (Step 7). That is the H_4 piece of BFKL⊗LO. If your BFKL list keeps $+\log\frac{\vee}{k^+}H_4$, Row IV must carry the overlap term, which removes the double counting:

$$-\log\frac{\vee}{k^+}H_4 = \underbrace{-\Big[\log\frac{\vee}{k^+}-1+\frac{k^+}{\vee}\Big]H_4}_{\text{your LO}^y\text{ term}}\ \underbrace{-\Big(1-\frac{k^+}{\vee}\Big)H_4}_{\text{your }H_4\text{ line}}. \tag{1.2}$$

  It is neither DGLAP (no pole, no μ) nor a genuine finite term.

- **The finite part of Row IV is only the linear-polarization term.**

- The JIMWLK term carries the ϵ-bracket $\mathcal{B}$, the $d_\perp$-dimensional transverse integral of this row; its pole cancels between the rows that make up JIMWLK⊗LO (`full_check`).

- $\mathrm{LO}|_{\alpha_s(\mu_R)}+\mathrm{RC}=\mathrm{LO}|_{\alpha_s\to\alpha_s(c_0/|y'-y|)}$, inside the integral. At leading power ($a\to 0$): $\int_a^1\to\int_0^1$.

All steps are checked in `checks.py`, `plus_check.py` (Sec. 5).

2 Notation

$d_\perp=2-\epsilon$; $c_0=2e^{-\gamma_E}$; $a\equiv\frac{k^+}{\vee}$; $\xi_\Lambda\equiv\frac{k^+}{k^++\Lambda}$; $\beta_0=\frac{11}{3}N_c$; $\frac{g^2N_c}{4\pi^2}=\frac{\alpha_s}{2\pi}2N_c$; $\frac{g^2}{8\pi^2}=\frac{\alpha_s}{2\pi}$.

$$\ell_\Delta(\mu)\equiv 2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}=\log\frac{(y'-y)^2\mu^2}{c_0^2}, \tag{2.1}$$

$$\begin{aligned}\mathrm{LO}_\xi[f]\equiv{}&\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,f(y'-y)\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\\ &\times\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x),\end{aligned} \tag{2.2}$$

$\mathrm{LO}\equiv\mathrm{LO}_{\xi=1}[1]$. The weight f is *inside* the integral.

- $\mathrm{LO}^{\mathrm{lp}}_\xi$: Eq. (2.2) with $f=1$ and the kernel of your (L3).

- $\mathrm{LO}^{\mathrm{sub}}_\xi[f]$: Eq. (2.2) with $\frac{(x'-y')}{(x'-y')^2}\to\frac{(x'-y)}{(x'-y)^2}$ and $U^{e'c}(y')\to U^{e'c}(y)$, i.e. the point y' is kept only in the phase and in f.

$$\mathrm{LO}^y \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x'}\frac{(x-y)\cdot(x'-y)}{(x-y)^2(x'-y)^2}\big[U^{e'c}(y)-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \qquad H_4 = \frac{g^2 N_c}{\pi\,\mathbf{k}^2}\,\mathrm{LO}^y, \tag{2.3}$$

H_4 being your term $\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}[\dots]$ $(y\leftrightarrow z)$: the w,w' integrals give $\frac{4\pi^2}{\mathbf{k}^2}$.

$$\mathcal{P}_{gg}(\xi) = 2N_c\Big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big]+\frac{\beta_0}{2}\delta(1-\xi), \qquad \int_a^1 d\xi\,\frac{\xi\,G(\xi)}{(1-\xi)_+}\Big|_{\rm std} \equiv \int_a^1 d\xi\frac{\xi G(\xi)-G(1)}{1-\xi}+G(1)\log(1-a). \tag{2.4}$$

3 The starting point

Your expression, in this notation (the ϵ-bracket is a function of $y'-y$, so it is a weight inside the integrals):

$$\begin{aligned}
(\mathrm{L1}) &= \frac{g^2N_c}{4\pi^2}\int_a^1\frac{d\xi}{\xi^2}\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\Big]\mathrm{LO}_\xi\big[\mathcal{B}_0\big],\\
(\mathrm{L2}) &= \frac{g^2N_c}{4\pi^2}\Big[\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big)\Big]\mathrm{LO}\big[\mathcal{B}_0\big],\\
(\mathrm{L3}) &= \frac{g^2N_c}{4\pi^2}\int_a^{\xi_\Lambda} d\xi\,\frac{1-\xi}{\xi}\,\mathrm{LO}^{\rm lp}_\xi,
\end{aligned} \tag{3.1}$$

with $\mathcal{B}_0 \equiv -\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}-1$.

Your plus prescription in (L1). With $p^+ = k^+\frac{1-\xi}{\xi}$, $\frac{d\xi}{\xi^2}\frac{\xi}{1-\xi}=\frac{dp^+}{p^+}$, and $\xi=1\leftrightarrow p^+=0$, $\xi=a\leftrightarrow p^+=\vee-k^+$. Your prescription is $\int_0^{\vee-k^+}dp^+\frac{F(p^+)-F(0)}{p^+}$, $F=\mathrm{LO}_\xi[\mathcal{B}_0]$, $F(0)=\mathrm{LO}[\mathcal{B}_0]$. The cut $p^+\geq\Lambda$ gives, exactly,

$$\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+}F = \int_0^{\vee-k^+}dp^+\frac{F(p^+)-F(0)}{p^+}+F(0)\Big[\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big)\Big]+O\Big(\frac{\Lambda}{k^+}\Big), \tag{3.2}$$

which is your (L2) (checked: 4×10^{-16}). In ξ, and compared with the standard plus distribution of Eq. (2.4) acting on F/ξ^2:

$$\int_a^1 d\xi\,\frac{F(\xi)-F(1)}{\xi(1-\xi)} = \int_a^1\frac{d\xi}{\xi^2}\frac{\xi}{(1-\xi)_+}\Big|_{\rm std}F(\xi)-F(1)\log\Big(\frac{\vee}{k^+}-1\Big) \qquad (\text{checked: } 6\times10^{-17}). \tag{3.3}$$

4 Step by step

Step 1: the ϵ-bracket. With $d_\perp=2-\epsilon$, $-\frac{d_\perp}{\epsilon}=-\frac{2}{\epsilon}+1$, so the -1 cancels and, exactly,

$$\mathcal{B}_0 = -\frac{2}{\epsilon}-\frac{d_\perp}{2}\Big[2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}\Big] = -\frac{2}{\epsilon}-\frac{d_\perp}{2}\ell_\Delta(\mu)\equiv\mathcal{B}. \tag{4.1}$$

Steps 2–3: (L1) with the standard plus distribution; the $\log(\frac{\vee}{k^+}-1)$ cancels. By Eq. (3.3), (L1) $=\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}P^{\rm r}(\xi)\mathrm{LO}_\xi[\mathcal{B}]-\frac{g^2N_c}{4\pi^2}\log\big(\frac{\vee}{k^+}-1\big)\mathrm{LO}[\mathcal{B}]$, with $P^{\rm r}\equiv\mathcal{P}_{gg}-\frac{\beta_0}{2}\delta(1-\xi)$ (standard plus distribution; $\frac{g^2N_c}{4\pi^2}=\frac{g^2}{8\pi^2}2N_c$). The second term cancels the same term of (L2):

$$(\mathrm{L1})+(\mathrm{L2}) = \frac{g^2N_c}{4\pi^2}\log\frac{k^+}{\Lambda}\mathrm{LO}[\mathcal{B}]+\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\,P^{\rm r}(\xi)\,\mathrm{LO}_\xi[\mathcal{B}]. \tag{4.2}$$

In your split this is the pair $+\frac{g^2N_c}{4\pi^2}\log\frac{\vee}{k^+}\mathrm{LO}[\mathcal{B}]-\frac{g^2N_c}{4\pi^2}\log\frac{\vee}{k^+}\mathrm{LO}[\mathcal{B}]$ (they cancel identically) plus the $\log(1-a)$ that completes the standard plus distribution.

Step 4: add and subtract $\frac{\beta_0}{2}\delta(1-\xi)$. At $\xi=1$, $\mathrm{LO}_\xi[f]=\mathrm{LO}[f]$:

$$\text{the }\xi\text{ integral of (4.2)} = \underbrace{\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\mathcal{P}_{gg}(\xi)\mathrm{LO}_\xi\Big[-\frac{2}{\epsilon}-\frac{d_\perp}{2}\ell_\Delta(\mu)\Big]}_{\text{(A)}}+\underbrace{\frac{g^2\beta_0}{16\pi^2}\mathrm{LO}\Big[\frac{2}{\epsilon}+\frac{d_\perp}{2}\ell_\Delta(\mu)\Big]}_{\text{(R)}}. \tag{4.3}$$

Step 5: DGLAP. The hadron is $\frac{d^3N_h}{d^3k}=\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^3}D_{h/g}(\zeta)\frac{d^3N_g}{d^3k}\big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\big)$. The $\overline{MS}$ fragmentation function is $D^{\rm bare}(\zeta)=D(\zeta,\mu_F)+\frac{g^2}{8\pi^2}\big[\frac{2}{\epsilon}+\log\frac{\mu^2}{\mu_F^2}\big]\int_\zeta^1\frac{dz}{z}\mathcal{P}_{gg}(z)D\big(\frac{\zeta}{z},\mu_F\big)$.

- Inserted into LO, the counterterm covers $a\leq\zeta\leq\zeta'\leq1$, the same region as the ξ integral of (A) ($\zeta=\zeta'\xi\geq a$). The standard plus distribution, endpoint $\log(1-a)$ included, is the one of the DGLAP kernel.

- So the pole of (A) cancels, and $\log \frac{\mu^2}{\mu_F^2}$ turns $\ell_\Delta(\mu)$ into $\ell_\Delta(\mu_F)$.
- The leftover $\frac{\epsilon}{2}\ell_\Delta$ of $\frac{d_\perp}{2}$ has no pole in front and vanishes.

$$(\mathrm{A}) \;\to\; -\frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big]. \tag{4.4}$$

Step 6: running coupling. LO is $\propto g^2$. The $\overline{MS}$ coupling counterterm, $g^2 \to g^2(\mu_R)\big[1-\frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}\big]$ ($\frac{2}{\epsilon} = \frac{1}{\epsilon_{D=4-2\epsilon}}$), adds $-\frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}\mathrm{LO}$. Then, at $\mu=\mu_R$ and with $d_\perp \to 2$,

$$(\mathrm{R}) - \frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}\mathrm{LO} = \frac{g^2\beta_0}{16\pi^2}\mathrm{LO}\Big[\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big], \tag{4.5}$$

and, with $\alpha_s(Q) = \alpha_s(\mu_R)\big[1-\frac{\alpha_s\beta_0}{4\pi}\log\frac{Q^2}{\mu_R^2}\big]$,

$$\mathrm{LO}\big|_{\alpha_s(\mu_R)} + (4.5) = \mathrm{LO}\big|_{\alpha_s\to\alpha_s(c_0/|y'-y|)} + O(\alpha_s^3). \tag{4.6}$$

Here y, y' are the points where the produced gluon crosses the target in the amplitude and in the conjugate amplitude. Through $e^{-ik\cdot(y'-y)}$, $|y'-y| \sim 1/|\mathbf{k}|$. The μ_R dependence cancels: $\partial_{\log\mu_R^2}$ of LO is $-\frac{\alpha_s^2\beta_0}{4\pi}\mathrm{LO}$, that of (4.5) is $+\frac{\alpha_s^2\beta_0}{4\pi}\mathrm{LO}$.

Step 7: the $\log\frac{\vee}{k^+}$ hidden in (4.4). For $\xi \to a \ll 1$ the phase $e^{-ik\cdot(y'-y)/\xi}$ forces $y' \to y$. The $\frac{1-\xi}{\xi}$ part of $\mathcal{P}_{gg}$, with $\frac{d\xi}{\xi^2}$, then behaves like $\frac{d\xi}{\xi}$ and gives $\log\frac{\vee}{k^+}$. Its coefficient is the $y'=y$ term, $\mathrm{LO}_\xi^{\mathrm{sub}}$. Add and subtract it:

$$(4.4) = \underbrace{-\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\Big\{\mathcal{P}_{gg}\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big] - 2N_c\frac{1-\xi}{\xi}\mathrm{LO}_\xi^{\mathrm{sub}}\big[\ell_\Delta(\mu_F)\big]\Big\}}_{\text{DGLAP, no }\log\frac{\vee}{k^+}} - \frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\,2N_c\frac{1-\xi}{\xi}\mathrm{LO}_\xi^{\mathrm{sub}}\big[\ell_\Delta(\mu_F)\big]. \tag{4.7}$$

In $\mathrm{LO}_\xi^{\mathrm{sub}}$ only the phase and the weight depend on y'. So:

- the constant and $\log\mu_F^2$ multiply $\int d^2y'\,e^{-ik\cdot(y'-y)/\xi} = (2\pi\xi)^2\delta^2(\mathbf{k}) = 0$ ($\mathbf{k}\neq 0$);
- $\int d^2y'\,e^{-ik\cdot(y'-y)/\xi}\log(y'-y)^2 = -\frac{4\pi\xi^2}{\mathbf{k}^2}$ (from $\nabla^2\log r^2 = 4\pi\delta^2(r)$; checked numerically).

Hence $\mathrm{LO}_\xi^{\mathrm{sub}}[\ell_\Delta(\mu_F)] = \mathrm{LO}_\xi^{\mathrm{sub}}[\log(y'-y)^2] = -\frac{4\pi\xi^2}{\mathbf{k}^2}\mathrm{LO}^y$, and

$$-\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}2N_c\frac{1-\xi}{\xi}\Big(-\frac{4\pi\xi^2}{\mathbf{k}^2}\Big)\mathrm{LO}^y = \frac{g^2N_c}{\pi\mathbf{k}^2}\mathrm{LO}^y\int_a^1 d\xi\frac{1-\xi}{\xi} = \Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]H_4. \tag{4.8}$$

- So the DGLAP term (4.4) contains $+\big[\log\frac{\vee}{k^+} - (1-a)\big]H_4$ from its lower end. $+\log\frac{\vee}{k^+}H_4$ is the H_4 piece of BFKL⊗LO.
- Keeping (4.4) whole (DGLAP) and $+\log\frac{\vee}{k^+}H_4$ in the BFKL list counts it twice; the overlap term (1.2) removes it. In your split it is your LO^y term plus your $-(1-\frac{k^+}{\vee})H_4$ line.

Step 8: (L3). $\int_a^{\xi_\Lambda} = \int_a^1 + O(\frac{\Lambda}{k^+})$: the integrand $\propto 1-\xi$ vanishes at $\xi=1$. At $\xi\to a$, $\mathrm{LO}_\xi^{\mathrm{lp}} = O(\xi^2)$ (the δ^{ij}-traceless part of the kernel has no $\delta^2(\mathbf{k})$ term), so there is no $\log\frac{\vee}{k^+}$. No pole multiplies it, so $d_\perp \to 2$:

$$(\mathrm{L3}) = \frac{g^2N_c}{4\pi^2}\int_a^1 d\xi\,\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^{\mathrm{lp}}\Big|_{d_\perp=2}. \tag{4.9}$$

Step 9: collect. Eq. (4.2) (JIMWLK + Steps 4–6) and Step 8 give Eq. (1.1): JIMWLK $\frac{g^2N_c}{4\pi^2}\log\frac{k^+}{\Lambda}\mathrm{LO}[\mathcal{B}]$; DGLAP (4.4); running coupling (4.5); finite: (L3). With the BFKL bookkeeping of your lists: $+\log\frac{\vee}{k^+}H_4$ (BFKL) and $-\log\frac{\vee}{k^+}H_4$ (overlap, Row IV).

5 Checks

check (`checks.py` → `checks.json`)	result
$\mathcal{B}_0 = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\ell_\Delta$ (sympy)	exact
$\frac{g^2N_c}{4\pi^2} = \frac{\alpha_s}{2\pi}2N_c$	exact
cut $p^+ \geq \Lambda$ = your p^+ plus $+F(0)[\log\frac{k^+}{\Lambda} + \log(\frac{\vee}{k^+}-1)]$ (`plus_check.py`)	4×10^{-16}
your p^+ plus = standard ξ plus $-F(1)\log(\frac{\vee}{k^+}-1)$, Eq. (3.3)	6×10^{-17}
$\int_a^1\frac{1-\xi}{\xi}d\xi = \log\frac{1}{a} - (1-a)$	exact
$2\pi\int rJ_0(qr)\log r^2 e^{-\delta r^2}dr \to -\frac{4\pi}{q^2}$ ($q=1.7$; $\delta = 10^{-2}, 3\times10^{-3}, 10^{-3}$)	-4.410, -4.366, $-4.354 \to -4.348$
$\frac{\alpha_s}{2\pi}2N_c\frac{4\pi}{\mathbf{k}^2} = \frac{g^2N_c}{\pi\mathbf{k}^2}$; H_4 normalisation (`g2iv_rc/compact`)	exact
running coupling after the counterterm; μ_R independence	exact